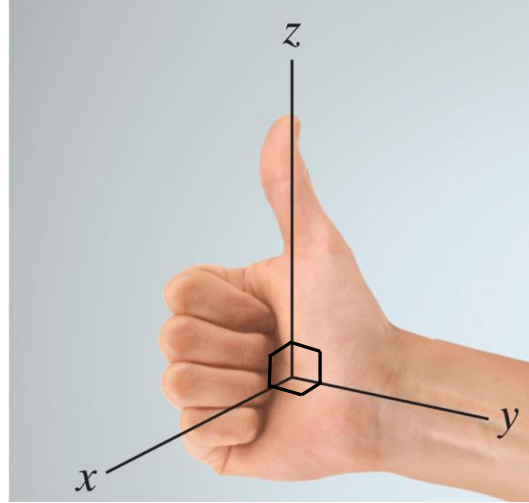


3D Force Vectors

- Right-Handed Coordinates System

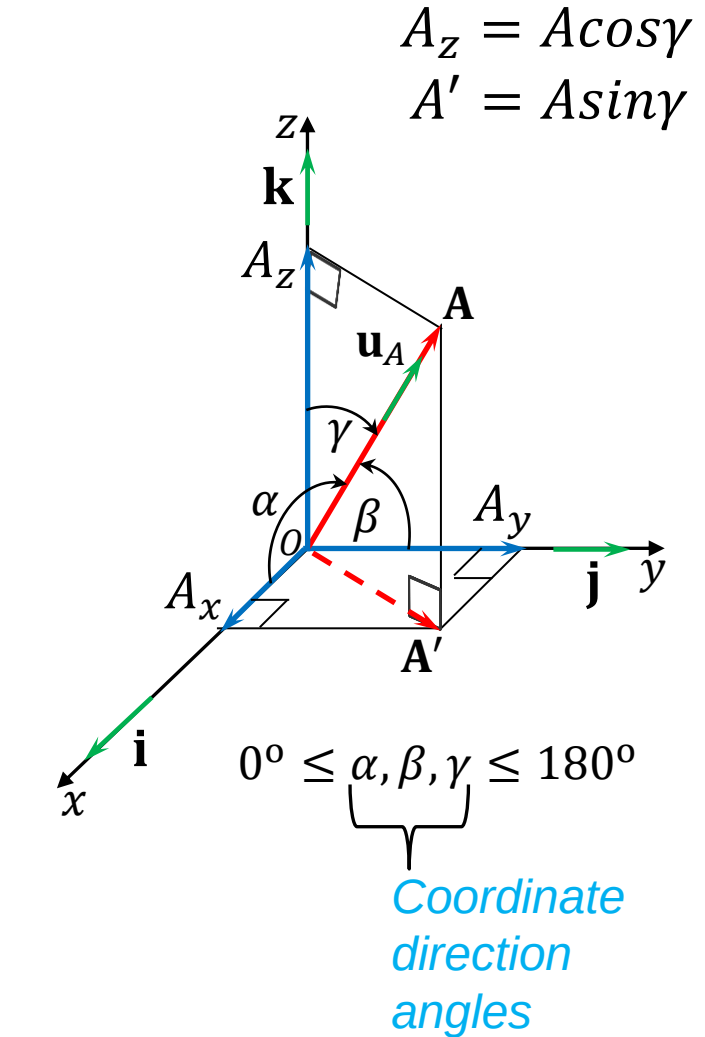


- 3D Cartesian Vector Representation

$$\mathbf{A} = \mathbf{A}' + A_z \mathbf{k} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

$$A = |\mathbf{A}| = \sqrt{(A_x)^2 + (A_y)^2 + (A_z)^2}$$

$$\mathbf{u}_A = \frac{\mathbf{A}}{A} = \frac{A_x}{A} \mathbf{i} + \frac{A_y}{A} \mathbf{j} + \frac{A_z}{A} \mathbf{k}$$



$$\mathbf{A} = A \mathbf{u}_A$$

3D Force Vectors

- Direction of a 3D Vector

- Direction cosines:

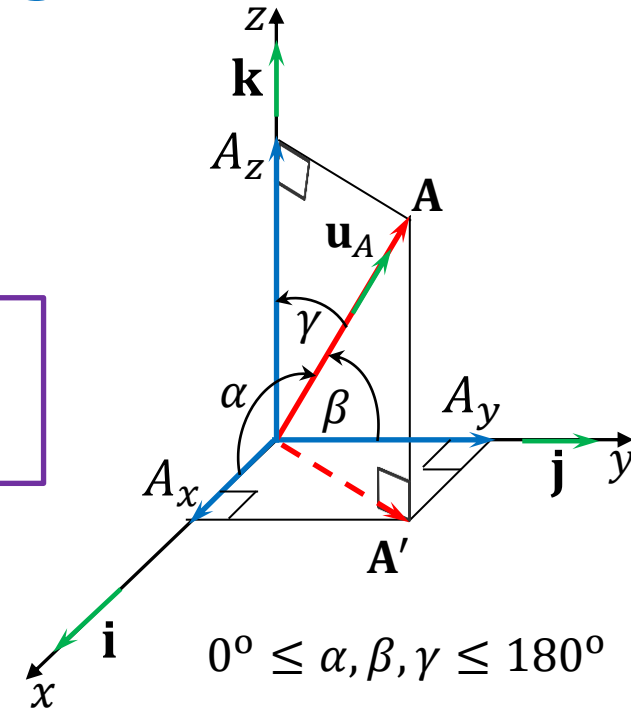
$$\cos \alpha = \frac{A_x}{A} \quad \cos \beta = \frac{A_y}{A} \quad \cos \gamma = \frac{A_z}{A}$$

$$\mathbf{u}_A = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

if $A_t > 0$, associate angle is acute

if $A_t < 0$, associate angle is obtuse

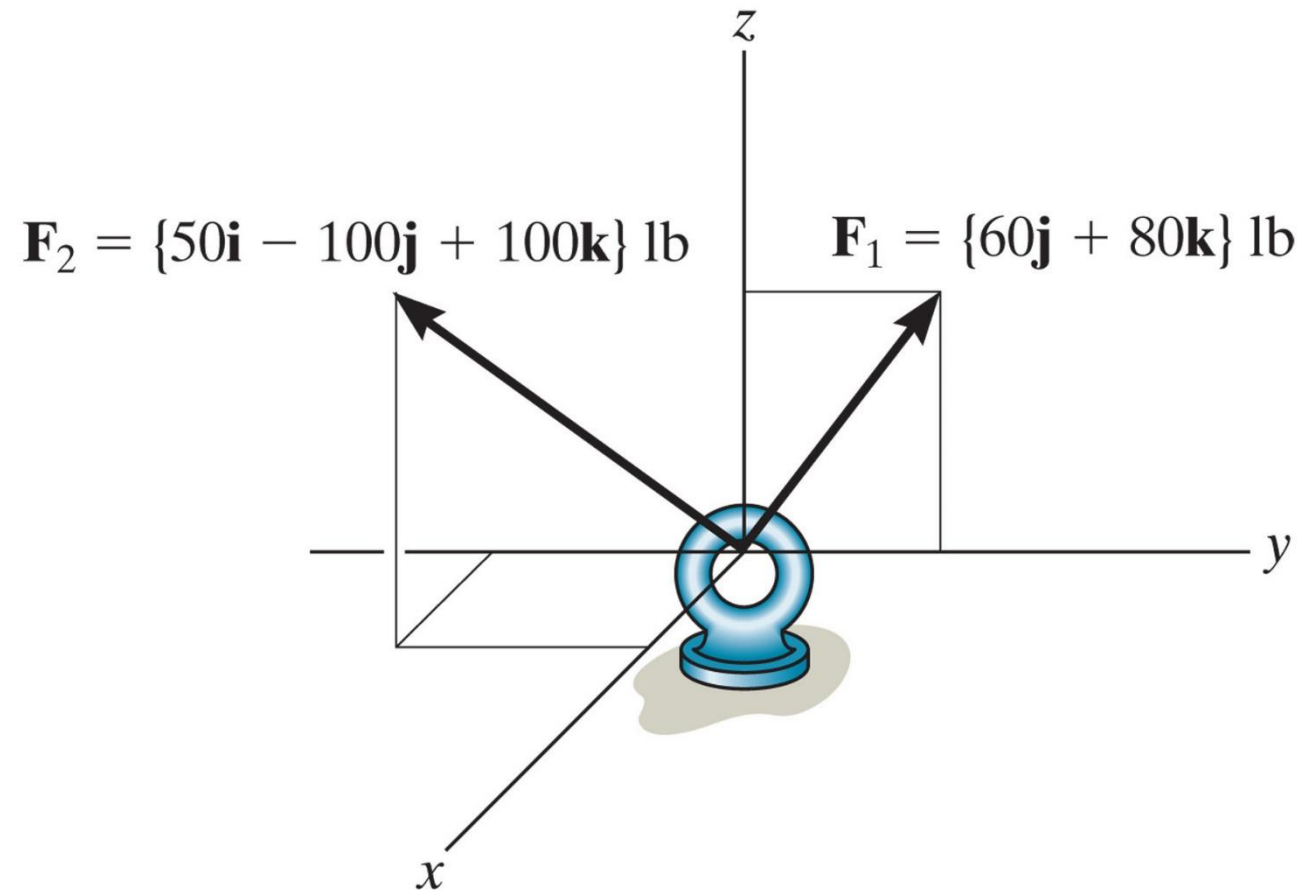


$t = x \text{ or } y \text{ or } z$

Example 8

Resultant of 3D Force Vectors

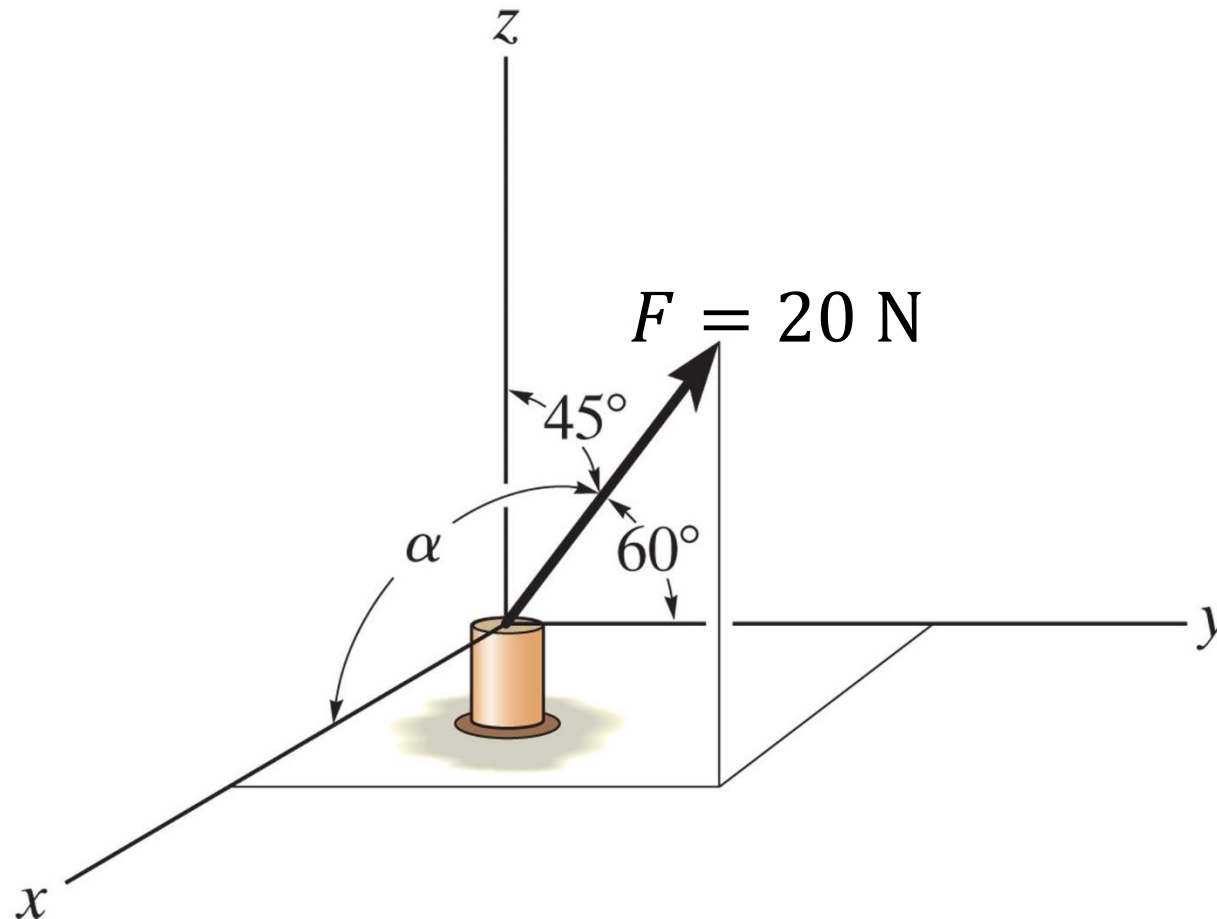
Determine the magnitude and the coordinate direction angles of the resultant force acting on the ring in the figure.



Example 9

- **Cartesian Force Vectors**

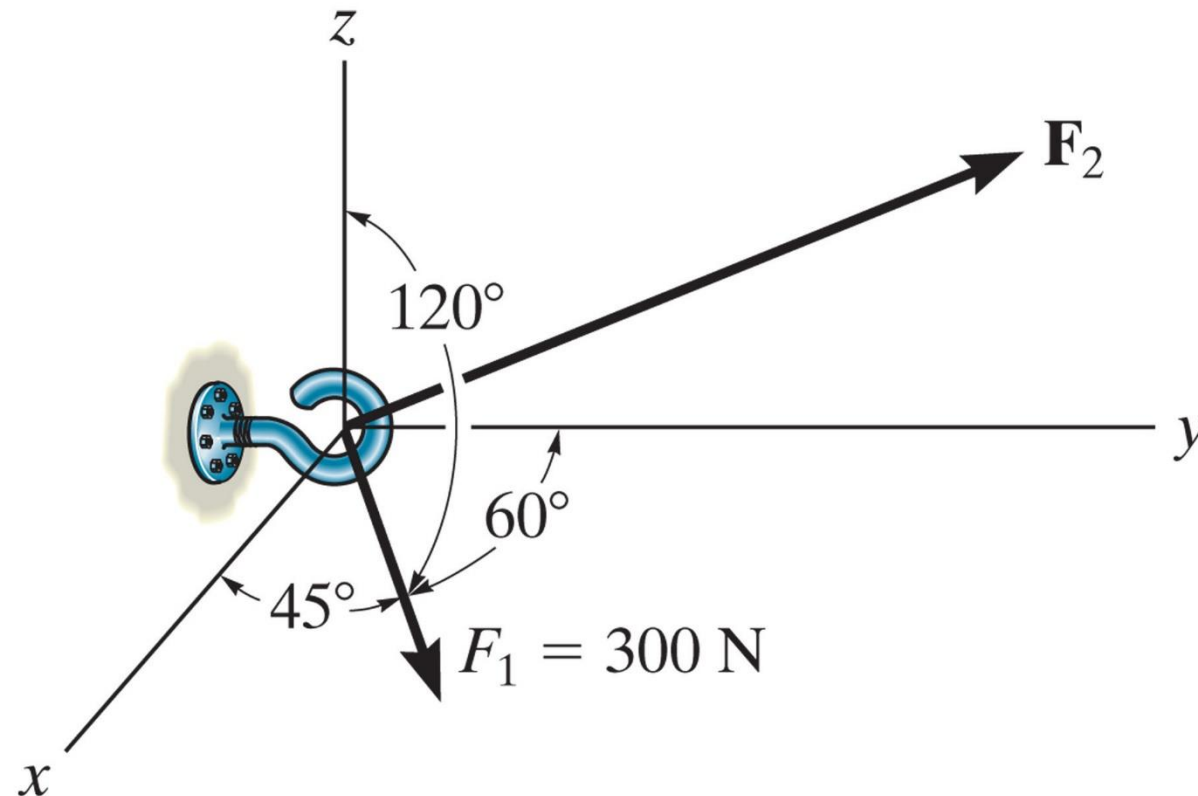
- (a) Determine the value of α .
- (b) Express the force $\mathbf{F}$ as a Cartesian vector.



Example 10

- **Addition of 3D Cartesian Force Vectors**

Two forces act on the hook shown in the figure. Specify the magnitude of $\mathbf{F}_2$ so that the resultant force $\mathbf{R}$ acts along the positive y -axis and has a magnitude of 1000 N.



Assignment #1

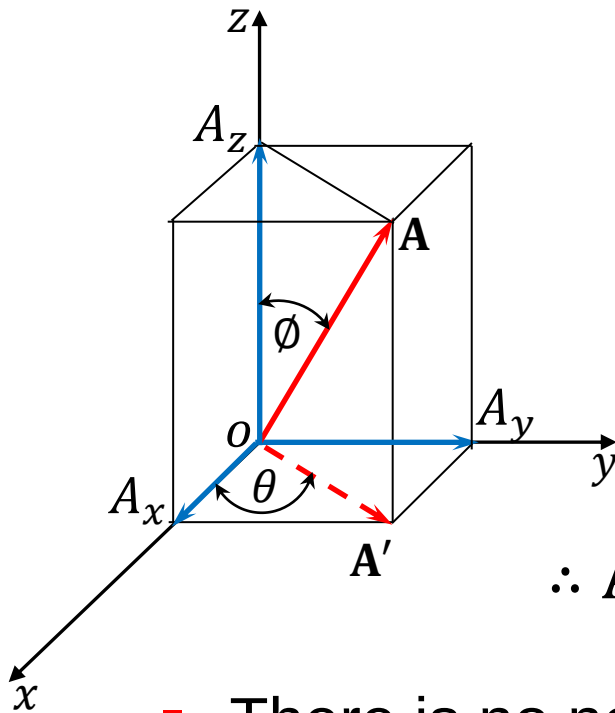
□ Coverage

- Parallelogram/Triangle Law
- Addition of a System of Coplanar Forces

□ **Due** Wednesday (21st February, 2024)

Force Vectors from other Projection Angles

- Determine a Vector from Transverse and Azimuthal Angles
 - Given vector A , azimuthal angle ϕ , and transverse angle θ
 - The two angles are not necessarily linked to α, β , and γ



$$A_z = A \cos \phi$$

$$A' = A \sin \phi$$

$$A_x = A' \cos \theta = A \sin \phi \cos \theta$$

$$A_y = A' \sin \theta = A \sin \phi \sin \theta$$

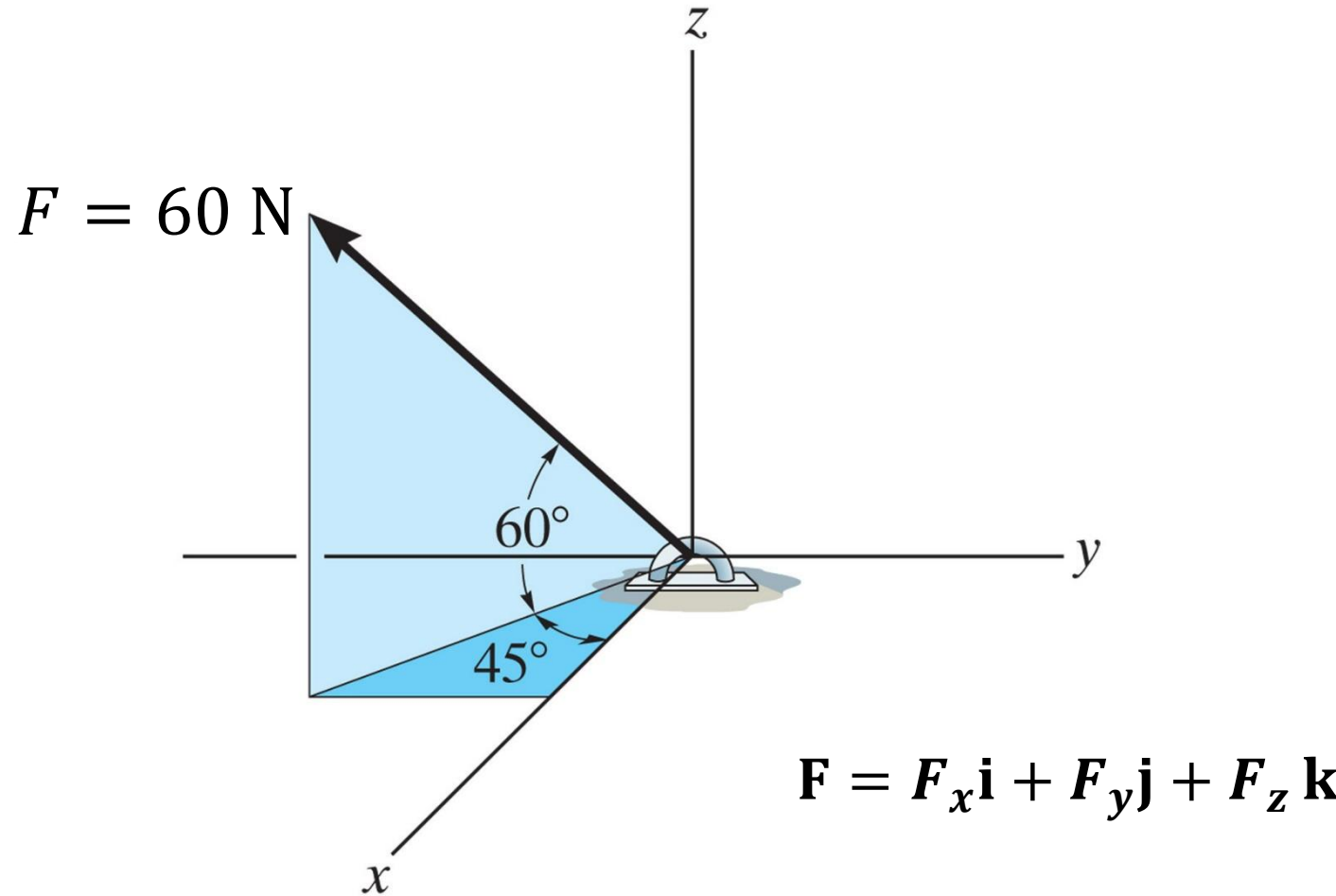
$$\therefore \mathbf{A} = A \sin \phi \cos \theta \mathbf{i} + A \sin \phi \sin \theta \mathbf{j} + A \cos \phi \mathbf{k}$$

- There is no need to memorize the above formula
- Should learn how to derive it using trigonometry

Example 11

- Cartesian Force Vectors from Projection Angles**

Express the force **F** shown in the figure as a Cartesian vector.



$$\mathbf{F} = 60\cos 60^\circ \cos 45^\circ \mathbf{i} - 60\cos 60^\circ \sin 45^\circ \mathbf{j} + 60\sin 60^\circ \mathbf{k}$$

Position Vectors

■ Position Vectors

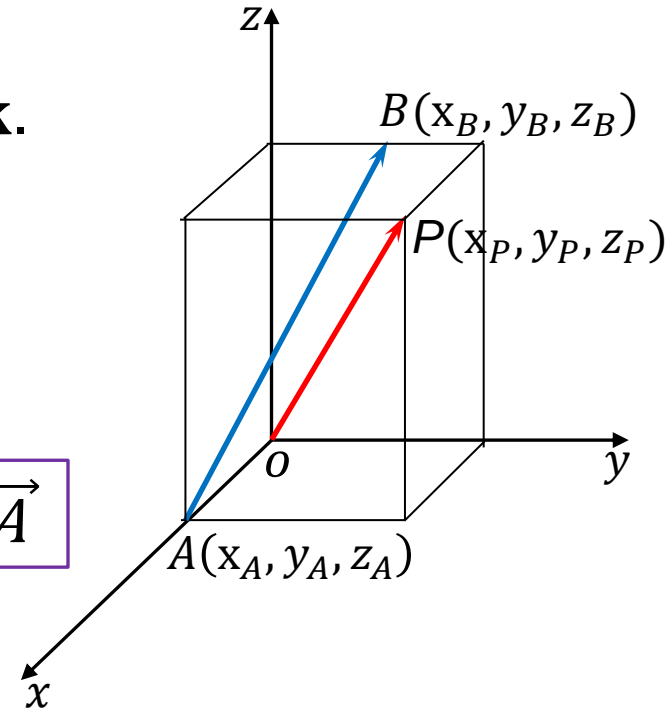
- Example: $\overrightarrow{AB}$, $\overrightarrow{OA}$, $\overrightarrow{OP}$, etc.
- It can be denoted as $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

$$\mathbf{r}_P = \overrightarrow{OP} = x_P\mathbf{i} + y_P\mathbf{j} + z_P\mathbf{k}$$

$$\mathbf{r}_{AB} = \mathbf{r}_B - \mathbf{r}_A$$

OR

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$



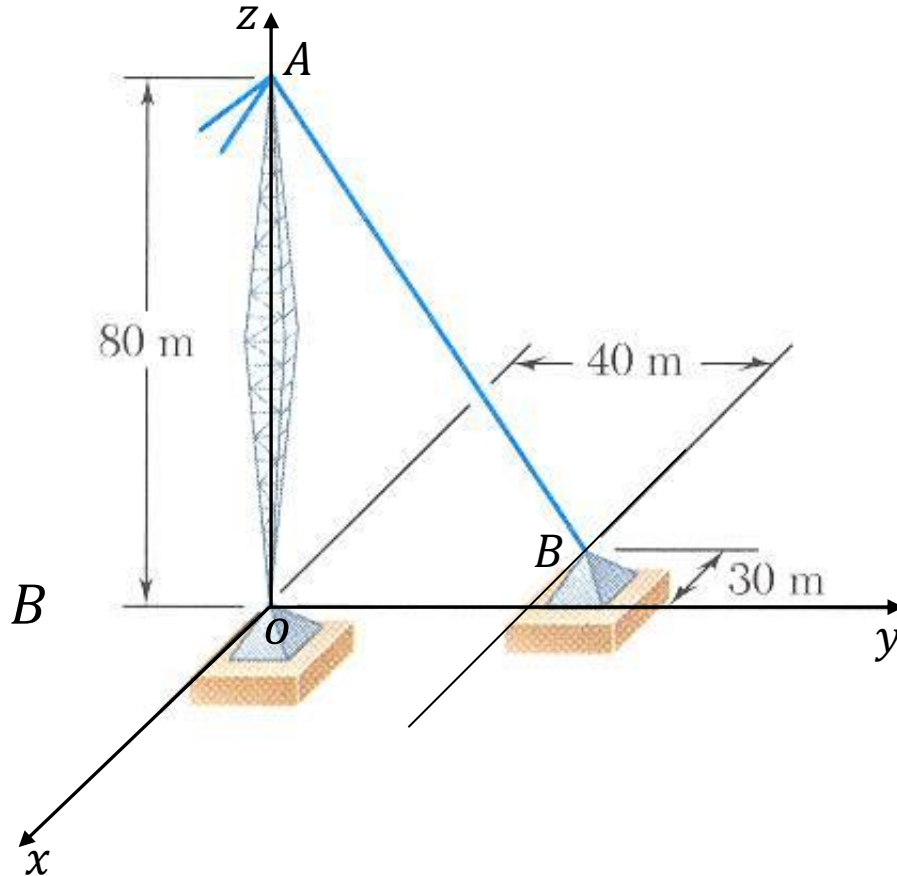
$$\mathbf{r}_{AB} = \overrightarrow{AB} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}$$

Example 12

■ Position Vectors

Determine:

- (a) $\overrightarrow{OA}$, $\overrightarrow{OB}$, and $\overrightarrow{AB}$
- (b) *length of the cable from A to B*
Give the exact answer.
- (c) *direction of the cable from A to B*
Give the exact answer.



The Unit Force Vector

- A force vector can be determined from its magnitude and direction specified by points on its line of action.

$$\mathbf{F} = F\mathbf{u}_{AB}$$

$$\mathbf{r}_{AB} = \mathbf{r}_B - \mathbf{r}_A$$

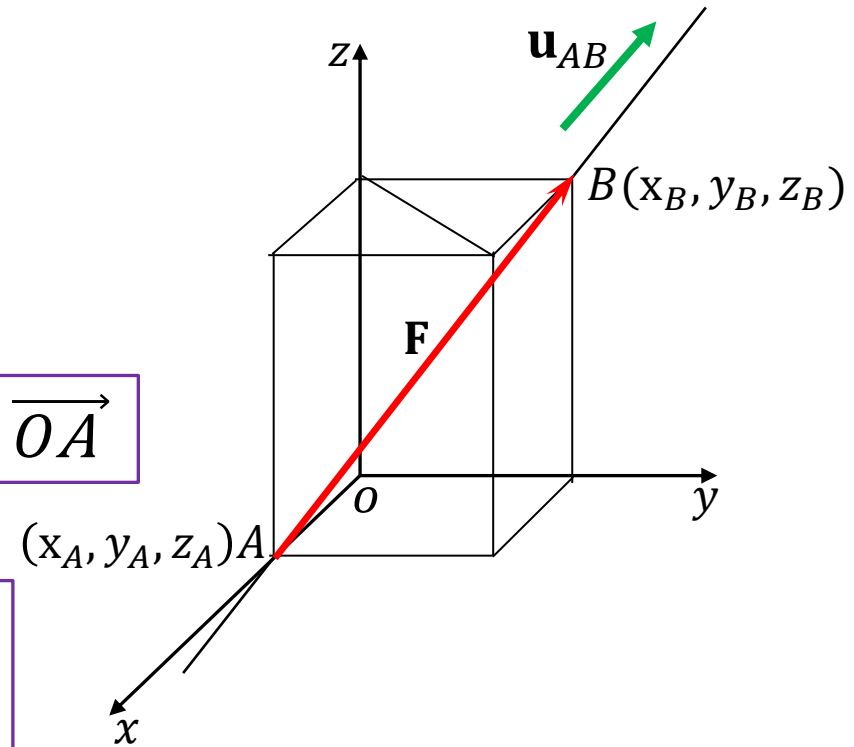
OR

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

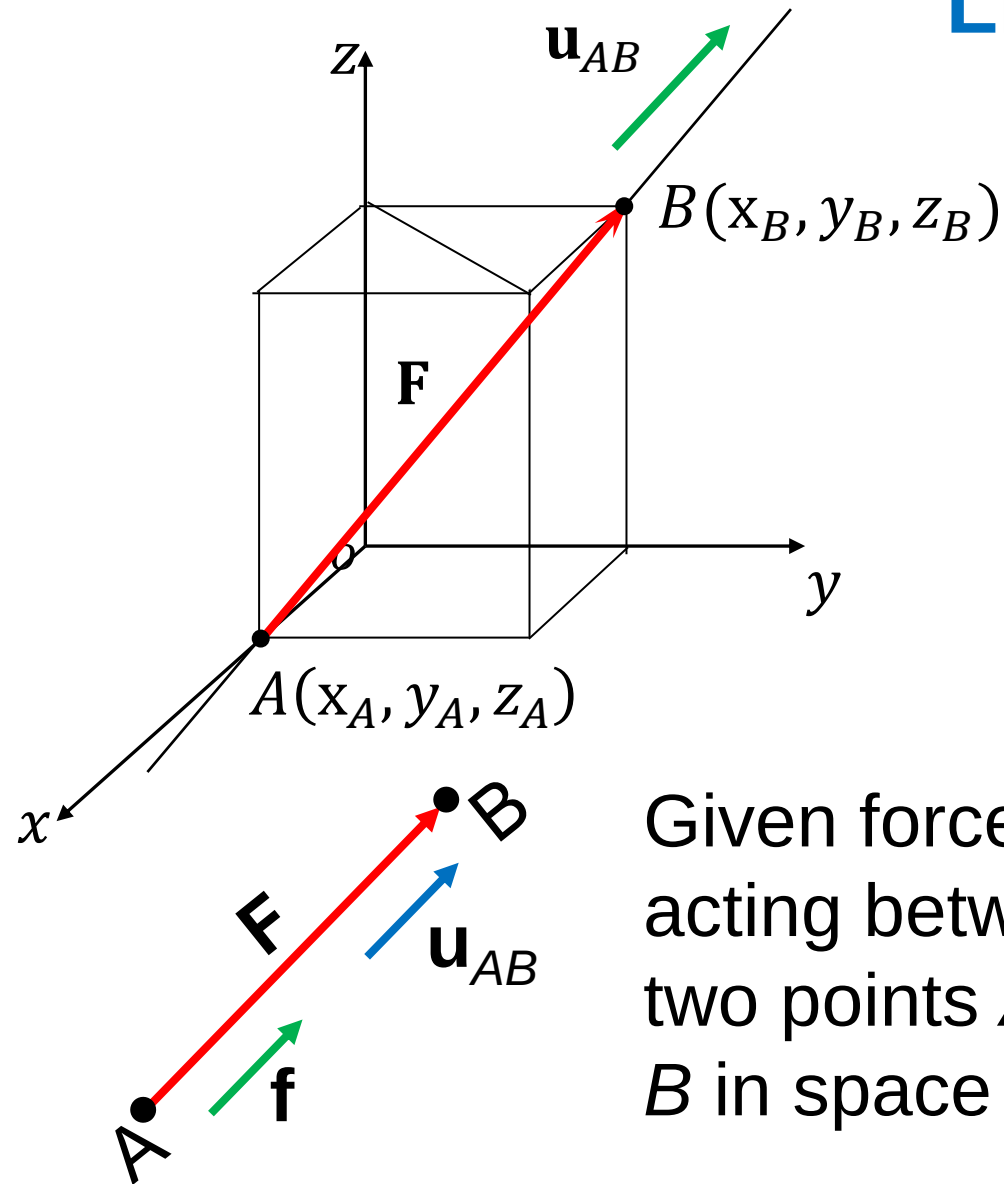
$$\mathbf{u}_{AB} = \frac{\mathbf{r}_{AB}}{|\mathbf{r}_{AB}|}$$

OR

$$\mathbf{u}_{AB} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$



Force Defined by its Magnitude and Spatial Unit Vector on its Line of Action



Given force, $\mathbf{F}$,
acting between
two points A and
B in space

Let $\mathbf{f}$ and $\mathbf{u}_{AB}$ be the unit force and unit spatial vectors, respectively, in the direction of force vector $\mathbf{F}$

$$\text{Then } \mathbf{f} = \frac{\mathbf{F}}{|\mathbf{F}|} = \frac{\mathbf{F}}{F} \quad \text{and} \quad \mathbf{u}_{AB} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

But by definition, $\mathbf{f} = \mathbf{u}_{AB}$

$$\therefore \frac{\mathbf{F}}{|\mathbf{F}|} = \mathbf{u}_{AB}$$

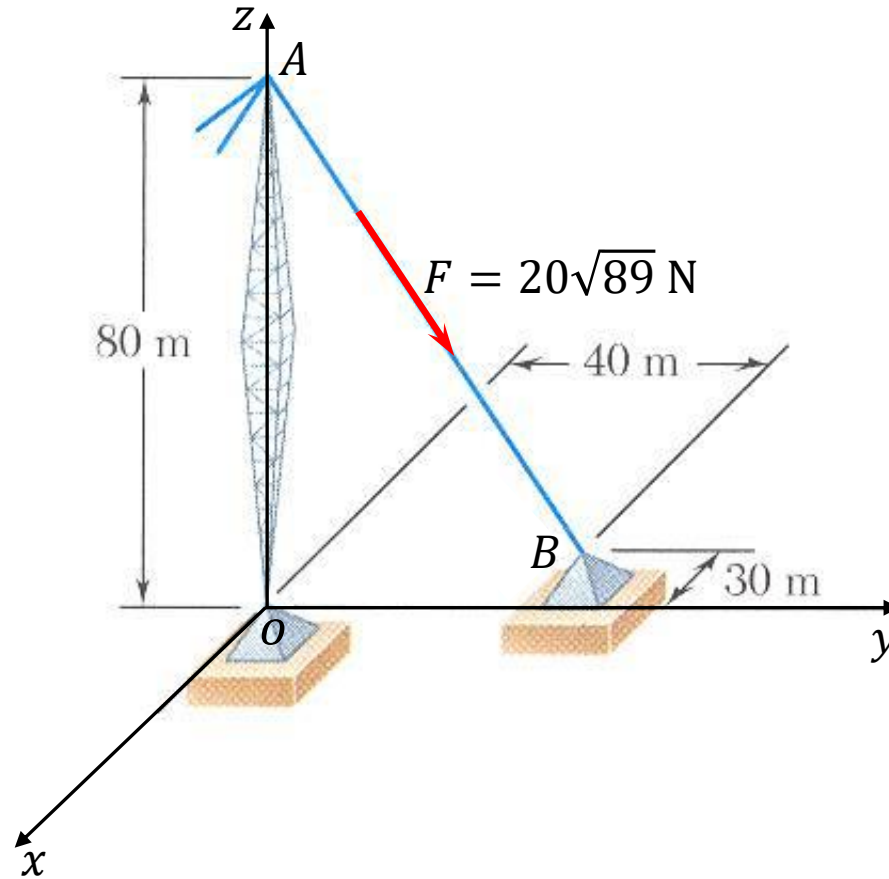
$$\Rightarrow \mathbf{F} = |\mathbf{F}| \mathbf{u}_{AB} \quad \text{or}$$

$$\boxed{\mathbf{F} = F \mathbf{u}_{AB}}$$

Recall, spatial vector $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

Example 16

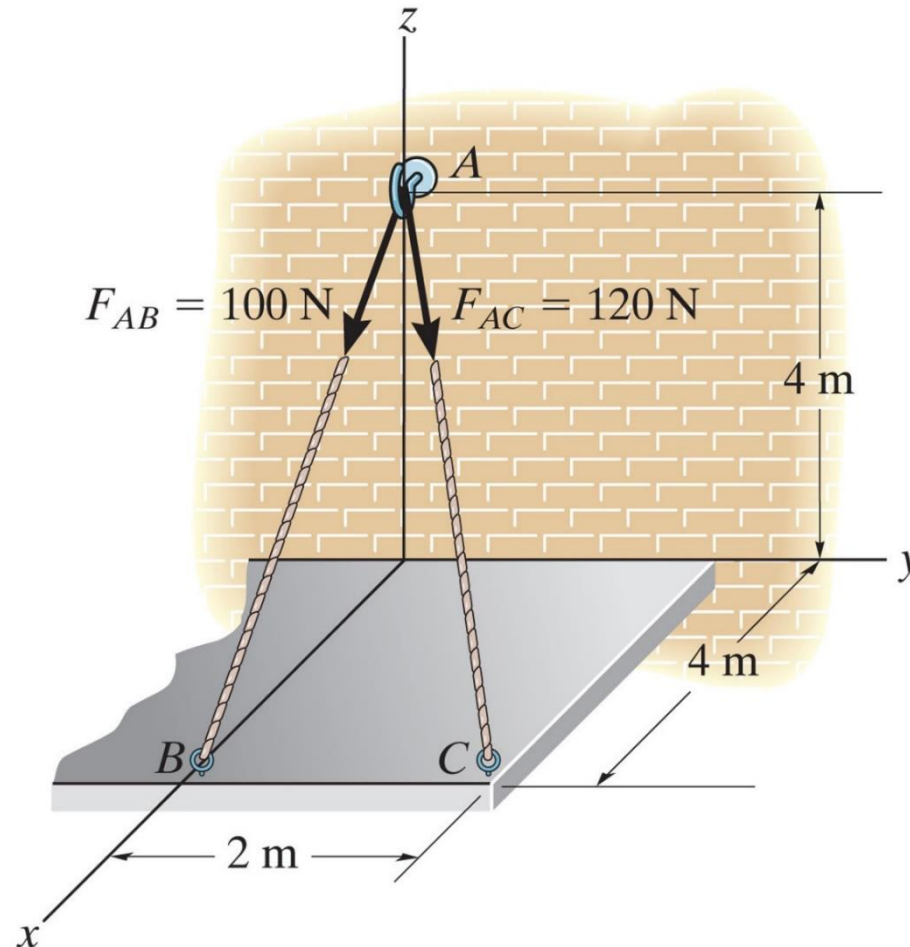
- If the tension in the cable AB is $20\sqrt{89}$ N, determine the force acting at A towards B.



Example 17

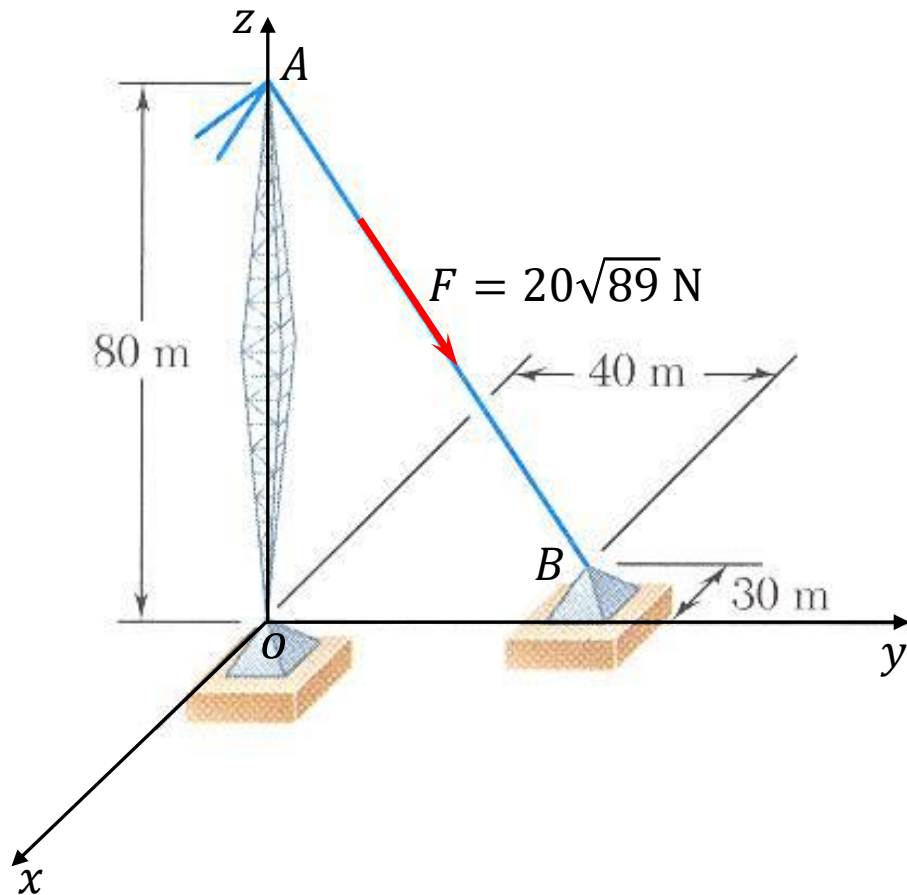
- Force Vector Directed Along a Line

The slab is supported by cables as shown in the shown. Determine the resultant force acting at A.



Example 16

- If the tension in the guy cable AB is $20\sqrt{89}$ N, determine the force acting at A towards B.



Solution

$$\mathbf{F} = |\mathbf{F}|\mathbf{u}_{AB}$$

$$\mathbf{u}_{AB} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$\overrightarrow{OB} = -30\mathbf{i} + 40\mathbf{j} + 0\mathbf{k} = (-30\mathbf{i} + 40\mathbf{j}) \text{ m}$$

$$\overrightarrow{OA} = 80\mathbf{k} \text{ m}$$

$$\overrightarrow{AB} = (-30\mathbf{i} + 40\mathbf{j} - 80\mathbf{k}) \text{ m}$$

$$\mathbf{u}_{AB} = \frac{-30\mathbf{i} + 40\mathbf{j} - 80\mathbf{k}}{\sqrt{[(-30)^2 + 40^2 + (-80)^2]}}$$

$$\mathbf{u}_{AB} = \frac{-30\mathbf{i} + 40\mathbf{j} - 80\mathbf{k}}{10\sqrt{89}}$$

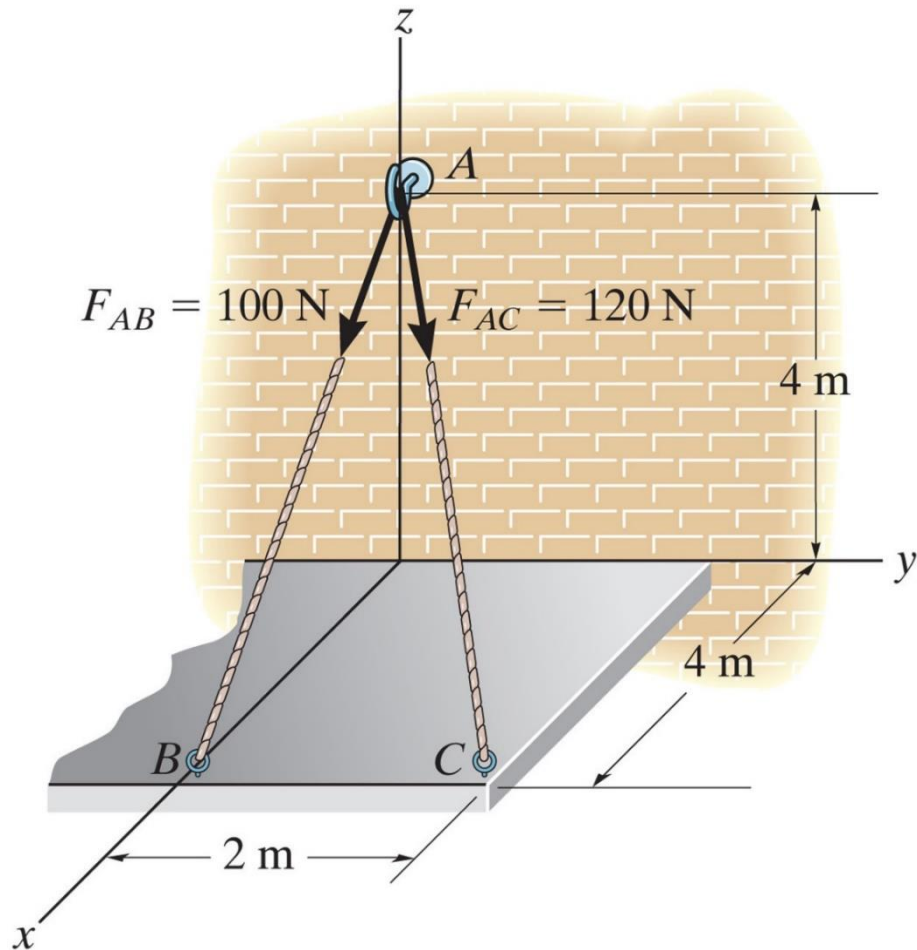
$$\mathbf{u}_{AB} = \frac{-3\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}}{\sqrt{89}}$$

$$\mathbf{F} = 20\sqrt{89} \times \frac{-3\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}}{\sqrt{89}}$$

$$\mathbf{F} = (-60\mathbf{i} + 80\mathbf{j} - 160\mathbf{k}) \text{ N}$$

Example 17

- The slab is supported by cables as shown in the diagram. Determine the resultant force acting at A.



Solution

$$\mathbf{F}_{AB} = F_{AB} \mathbf{u}_{AB}$$

$$\mathbf{F}_{AC} = F_{AC} \mathbf{u}_{AC}$$

$$\mathbf{u}_{AB} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

$$\mathbf{u}_{AC} = \frac{\overrightarrow{AC}}{|\overrightarrow{AC}|}$$

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= 4\mathbf{i} - 4\mathbf{k} \end{aligned}$$

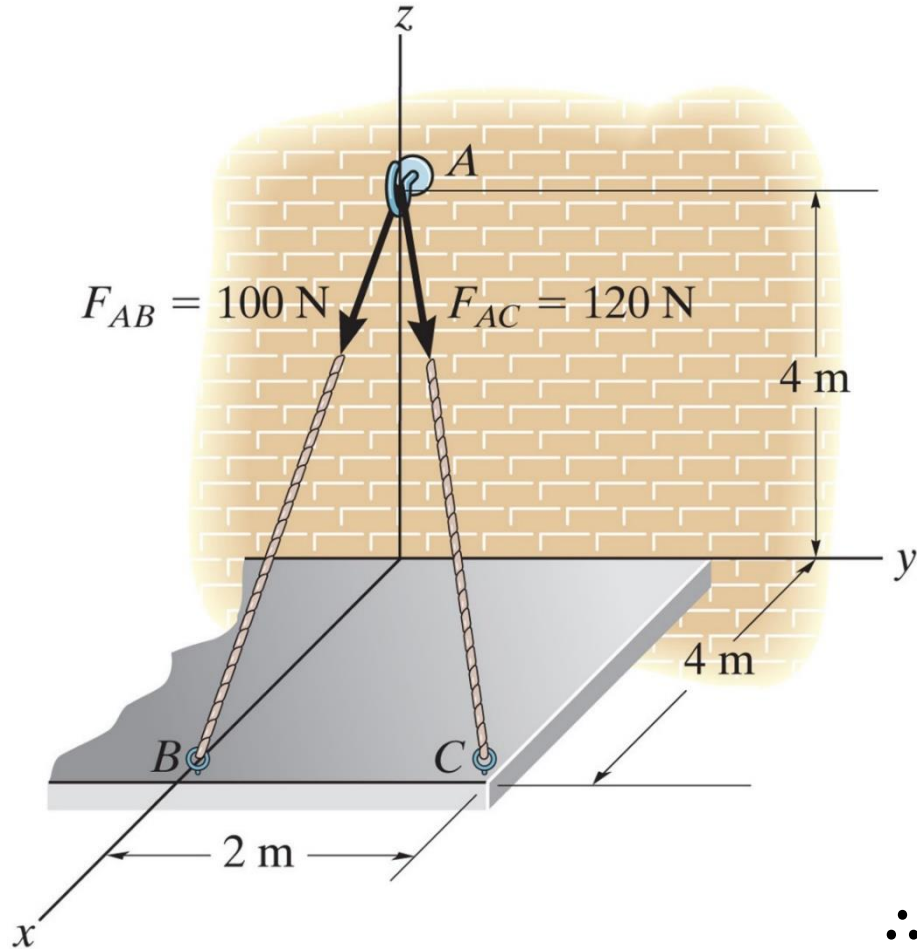
$$\mathbf{u}_{AB} = \frac{4\mathbf{i} - 4\mathbf{k}}{\sqrt{[4^2 + (-4)^2]}}$$

$$= 0.707\mathbf{i} - 0.707\mathbf{k}$$

$$\begin{aligned} \overrightarrow{AC} &= \overrightarrow{OC} - \overrightarrow{OA} \\ &= (4\mathbf{i} + 2\mathbf{j}) - 4\mathbf{k} \\ &= 4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k} \end{aligned}$$

$$\begin{aligned} \mathbf{u}_{AC} &= \frac{4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}}{\sqrt{[4^2 + 2^2 + (-4)^2]}} \\ &= 0.667\mathbf{i} + 0.333\mathbf{j} - 0.667\mathbf{k} \end{aligned}$$

Example 17 – Solution Cont'd



$$\begin{aligned}\mathbf{F}_{AB} &= F_{AB}\mathbf{u}_{AB} \\ &= 100(0.707\mathbf{i} - 0.707\mathbf{k}) \\ &= (70.7\mathbf{i} - 70.7\mathbf{k}) \text{ N}\end{aligned}$$

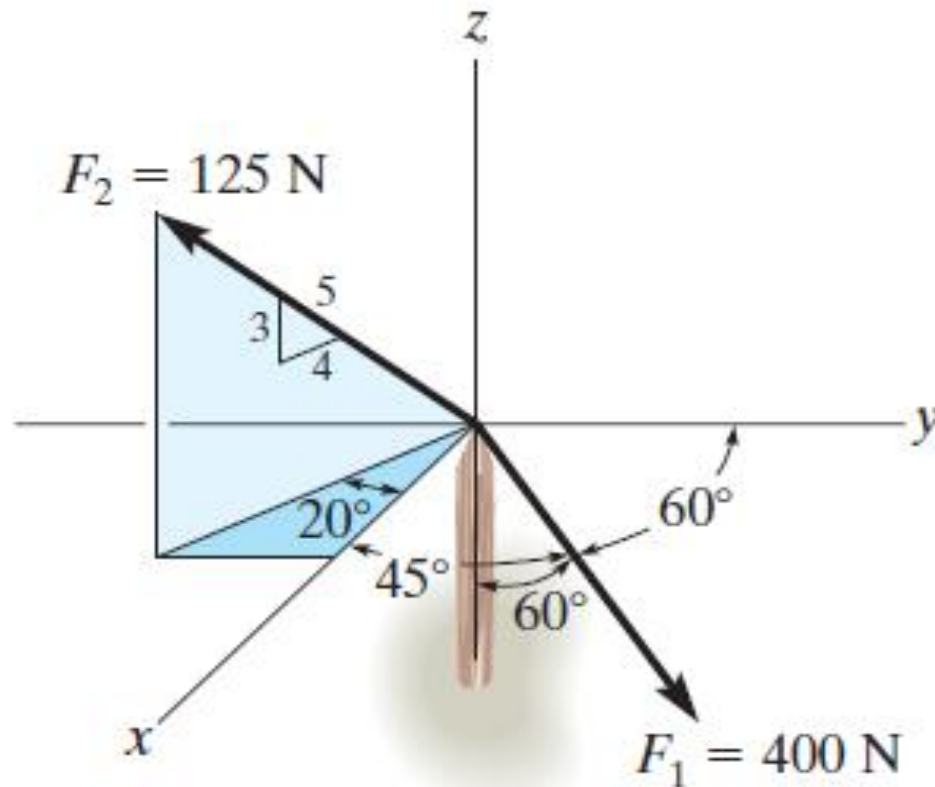
$$\begin{aligned}\mathbf{F}_{AC} &= F_{AC}\mathbf{u}_{AC} \\ &= 120(0.667\mathbf{i} + 0.333\mathbf{j} - 0.667\mathbf{k}) \\ &= (80.04\mathbf{i} + 39.96\mathbf{j} - 80.04\mathbf{k}) \text{ N}\end{aligned}$$

$$\text{Resultant force, } \mathbf{F}_R = \mathbf{F}_{AB} + \mathbf{F}_{AC}$$

$$\begin{aligned}\therefore \mathbf{F}_R &= \mathbf{F}_{AB} + \mathbf{F}_{AC} \\ &= (70.7\mathbf{i} - 70.7\mathbf{k}) + (80.04\mathbf{i} + 39.96\mathbf{j} - 80.04\mathbf{k}) \\ &= (150.74\mathbf{i} + 39.96\mathbf{j} - 150.74\mathbf{k}) \text{ N}\end{aligned}$$

Sample Problems

- 1) Determine the magnitude and coordinate direction angles of the resultant of the two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ in the diagram below.



Sample Problems

- 2) Express both of the forces $\mathbf{F}_{AB}$ and $\mathbf{F}_{AC}$ in Cartesian vector form and determine the magnitude and coordinate direction angles of the resultant force.

